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Vertical tire changer with double-sided tool and overturning structure

Abstract

A vertical tire changer with a double-sided tool and an overturning structure is provided. The vertical tire changer includes a pedal, a base, an operating platform, a pulley, a double-sided tool rotating structure, a small pressing wheel structure, a large pressing wheel structure, a lifting seat and a rotating chuck. A wheel hub is clamped above the rotating chuck so that the rotating chuck rotates to a horizontal position. The ends of the double-sided tool rotating structure are provided with a straight hook portion and a bent hook portion respectively. The straight hook portion is configured for tire mounting, and the bent hook portion is configured for tire dismounting. The small pressing wheel structure and the large pressing wheel structure can both be contracted and extended in the using process. The small pressing wheel structure and the large pressing wheel structure play an auxiliary supporting role.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application claims the benefit and priority of Chinese Patent Application No. 202210649163.7 filed with the China National Intellectual Property Administration on Jun. 9, 2022, the disclosure of which is incorporated by reference herein in its entirety as part of the present application.

BACKGROUND

Related Field

(2) The present disclosure relates to the technical field of tire mounting and dismounting, in particular to a vertical tire changer with a double-sided tool and an overturning structure.

Related Art

(3) Particularly, in the international market, the tire changer has come into families and is a necessary household appliance for families. Almost each of families in the foreign countries has not only a small automobile but also a large automobile.

(4) Due to the fact that various steel wire tires, explosion-proof tires and special wheel hubs appear along with the development of the automobile industry, a traditional pneumatic tire changer cannot meet the use requirement of various tires.

BRIEF SUMMARY

(5) The present disclosure aims to provide a vertical tire changer with a double-sided tool and an overturning structure and solves the technical problem that a traditional pneumatic tire changer cannot meet the use requirements of various tires due to the occurrence of various steel wire tires, explosion-proof tires and special wheel hubs along with the development of the automobile industry.

(6) In order to achieve the above-mentioned purpose, the present disclosure provides a vertical tire changer with a double-sided tool and an overturning structure. The vertical tire changer includes a pedal, a base, an operating platform, a sliding body, a double-sided tool rotating structure, a small pressing wheel structure, a large pressing wheel structure, a lifting seat and a rotating chuck. The operating platform is arranged on a first side of the base. The pedal is arranged beneath the operating platform. The sliding body is arranged above the base. The lifting seat is arranged above the base. The sliding body is arranged on one side of the lifting seat. The double-sided tool rotating structure is mounted on one side of the sliding body. The small pressing wheel structure and the large pressing wheel structure are mounted on an other side of the sliding body. The small pressing wheel structure is located above the large pressing wheel structure. The rotating chuck is arranged on a second side of the base, and the pedal is configured for controlling the rotating chuck.

(7) In some embodiments, a first mounting hole may be formed in the one side of the sliding body, and a second mounting hole and a third mounting hole may be formed in the other side of the sliding body.

(8) In some embodiments, the double-sided tool rotating structure may include a first rod body, a cylinder lug, a pin seat, a handball, a locking pin shaft, a double-sided tool and a pin. The first rod body may be mounted on the one side of the sliding body through the first mounting hole. The pin seat may be arranged at one end of the first rod body. The locking pin shaft may be arranged inside the pin seat. The handball may be arranged on a top of the locking pin shaft, and the pin may be arranged between the pin seat and the first rod body.

(9) In some embodiments, the double-sided tool may include a rotating shaft, set screws, a tool head and a hexagon socket head cap screw. The rotating shaft may be arranged inside the pin seat. The tool head may be detachably connected with the rotating shaft through the hexagon socket head cap screw, and the set screws may be arranged between the tool head and the rotating shaft.

(10) In some embodiments, the double-sided tool further may include a first gasket and a second gasket, and the first gasket and the second gasket may be arranged at two ends of the tool head respectively.

(11) In some embodiments, the small pressing wheel structure may include a nylon wheel and a pressing rod. One end of the pressing rod may be mounted on the one side of the sliding body through the second mounting hole, and the nylon wheel may be mounted at an other end of the pressing rod.

(12) In some embodiments, the lifting seat may include a slideway and a cylinder rod. The slideway may be fixedly connected with the base and located above the base. The sliding body may

be slidably connected with the slideway and located on one side of the slideway. The cylinder rod may be arranged on an other side of the slideway, and the cylinder rod may be connected with the sliding body.
(13) According to the vertical tire changer with a double-sided tool and an overturning structure provided by the present embodiment, a wheel hub is clamped above the rotating chuck so that the rotating chuck rotates to a horizontal position. The ends of the double-sided tool rotating structure are provided with a straight hook portion and a bent hook portion respectively. The straight hook portion is configured for tire mounting, and the bent hook portion is configured for tire dismounting. The small pressing wheel structure and the large pressing wheel structure can both be contracted and extended in the using process. The small pressing wheel structure and the large pressing wheel structure play an auxiliary supporting role. The rotation of the wheel hub is achieved through the rotating chuck. The lifting seat is pressed downwards, so that tire mounting or tire dismounting is completed. The speeds of both the tire mounting and the tire dismounting are increased, the types of tires that are mounted and dismounted are increased. Thus, this vertical tire changer can be dual-purpose, so the equipment cost is saved, and damages to wheel hubs and tires are reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to more clearly illustrate the technical solutions of the present disclosure or the prior art, the drawings, which need to be used in the embodiments or the prior art description, are briefly described below.
(2) FIG. **1** is a schematic diagram of an integral structure of a vertical tire changer with a double-sided tool and an overturning structure according to embodiments of the present disclosure.
(3) FIG. **2** is a decomposition schematic diagram of a vertical tire changer with a double-sided tool and an overturning structure according to embodiments of the present disclosure.
(4) FIG. **3** is a mounting schematic diagram of a sliding body according to embodiments of the present disclosure;
(5) FIG. **4** is a schematic structural diagram of a sliding body according to embodiments of the present disclosure;
(6) FIG. **5** is a section view of a double-sided tool rotating structure according to embodiments of the present disclosure;
(7) FIG. **6** is an explosive view of a double-sided tool according to embodiments of the present disclosure;
(8) FIG. **7** is a decomposition schematic diagram of a double-sided tool rotating structure according to embodiments of the present disclosure;
(9) FIG. **8** is a schematic structural diagram of a small pressing wheel structure according to embodiments of the present disclosure; and
(10) FIG. **9** is a section view of a rotating chuck according to embodiments of the present disclosure.
DETAILED DESCRIPTION OF VARIOUS EMBODIMENTS
(11) Embodiments of the present disclosure are described in detail below, and examples of the embodiments are illustrated in the attached figures. The embodiments described below by reference to the attached figures are exemplary only for explaining the present disclosure and are not to be construed as limiting the present disclosure.
(12) List of the reference characters: **1** pedal; **2** base; **3** operating platform; **4** sliding body; **5** double-sided tool rotating structure; **6** small pressing wheel structure; **7** large pressing wheel structure; **8** rotating chuck; **9** first mounting hole; **10** second mounting hole; **11** third mounting

hole; **12** first rod body; **13** cylinder lug; **14** pin seat; **15** handball; **16** locking pin shaft; **17** double-sided tool; **18** pin; **19** rotating shaft; **20** set screw; **21** tool head; **22** hexagon socket head cap screw; **23** nylon wheel; **24** pressing rod; **25** slideway; **26** cylinder rod; **27** pushing arm; **28** rotating seat; **29** fixed mount; **30** driving motor; **31** rotating disc; **32** clamping jaw; **33** second cylinder; **34** ejector rod; **35** first rotating rod; **36** seat body; **37** second rotating rod; **38** frame body; **39** fixed rod; **40** first gasket; and **41** second gasket.

(13) Referring to FIG. **1** to FIG. **3**, the present embodiments provide a vertical tire changer with a double-sided tool **17** and an overturning structure. The vertical tire changer includes a pedal **1**, a base **2**, an operating platform **3**, a sliding body **4**, a double-sided tool rotating structure **5**, a small pressing wheel structure **6**, a large pressing wheel structure **7**, a lifting seat and a rotating chuck **8**. The operating platform **3** is arranged on a first side of the base **2**. The pedal **1** is arranged beneath the operating platform **3**. The sliding body **4** is arranged above the base **2**. The lifting seat is arranged above the base **2**. The sliding body **4** is arranged on one side of the lifting seat. The double-sided tool rotating structure **5** is mounted on one side of the sliding body **4**. The small pressing wheel structure **6** and the large pressing wheel structure **7** are mounted on an other side of the sliding body **4**. The small pressing wheel structure **6** is located above the large pressing wheel structure **7**. The rotating chuck **8** is arranged on a second side of the base **2**, and the pedal **1** is configured for controlling the rotating chuck **8**.

(14) In the embodiment, a wheel hub is clamped above the rotating chuck **8** so that the rotating chuck **8** rotates to a horizontal position. The two ends of the double-sided tool rotating structure **5** are provided with a straight hook portion and a bent hook portion respectively. The straight hook portion is configured for tire mounting, and the bent hook portion is configured for tire dismounting. The small pressing wheel structure **6** and the large pressing wheel structure **7** can both be contracted and extended in the using process. The small pressing wheel structure **6** and the large pressing wheel structure **7** play an auxiliary supporting role. The rotation of the wheel hub is achieved through the rotating chuck **8**. The lifting seat is pressed downwards, so that tire mounting or tire dismounting is completed. The speeds of both the tire mounting and dismounting are increased, the types of tires that are mounted and dismounted are increased. Thus, this vertical tire changer can be dual-purpose, so the equipment cost is saved, and damages to wheel hubs and tires are reduced.

(15) A first mounting hole **9** is formed in the one side of the sliding body **4**, and a second mounting hole **10** and a third mounting hole **11** are formed in the other side of the sliding body **4**.

(16) In the embodiment, the first mounting hole **9** is configured for mounting the double-sided tool rotating structure **5**, the second mounting hole **10** is configured for mounting the small pressing wheel structure **6**, and the third mounting hole **11** is configured for mounting the large pressing wheel structure **7**.

(17) The double-sided tool rotating structure **5** includes a first rod body **12**, a cylinder lug **13**, a pin seat **14**, a handball **15**, a locking pin shaft **16**, a double-sided tool **17** and a pin **18**. The first rod body **12** is mounted on the one side of the sliding body **4** through the first mounting hole **9**. The pin seat **14** is arranged at one end of the first rod body **12**. The locking pin shaft **16** is arranged inside the pin seat **14**. The handball **15** is arranged on a top of the locking pin shaft **16**, and the pin **18** is arranged between the pin seat **14** and the first rod body **12**. The double-sided tool **17** includes a rotating shaft **19**, set screws **20**, a tool head **21** and a hexagon socket head cap screw **22**. The rotating shaft **19** is arranged inside the pin seat **14**. The tool head **21** is detachably connected with the rotating shaft **19** through the hexagon socket head cap screw **22**, and the set screws **20** are arranged between the tool head **21** and the rotating shaft **19**. The double-sided tool **17** further includes a first gasket **40** and a second gasket **41**, and the first gasket **40** and the second gasket **41** are arranged at the two ends of the tool head **21** respectively.

(18) In the embodiment, the rotation of the rotating shaft **19** is limited through the handball **15** and the locking pin shaft **16**. One end of the tool head **21** is a bent hook portion, and the other end of

the tool head **21** is a straight hook portion. A cylinder is arranged in the sliding body **4**, so that the first rod body **12** is controlled to stretch and retract along the first mounting hole **9**.

(19) The small pressing wheel structure **6** includes a nylon wheel **23** and a pressing rod **24**. One end of the pressing rod **24** is mounted on one side of the sliding body **4** through the second mounting hole **10**, and the nylon wheel **23** is mounted at the other end of the pressing rod **24**.

(20) In the embodiment, by arranging the nylon wheel **23**, the friction between the nylon wheel **23** and the tire is increased, and the tie is convenient to be dismounted or mounted.

(21) The lifting seat includes a slideway **25** and a cylinder rod **26**. The slideway **25** is fixedly connected with the base **2** and located above the base **2**. The sliding body **4** is slidably connected with the slideway **25** and located on one side of the slideway. The cylinder rod **26** is arranged on the other side of the slideway **25**, and the cylinder rod **26** is connected with the sliding body **4**.

(22) In the embodiment, the cylinder rod **26** pushes the sliding body **4** to move up and down along the slideway **25**, so that the positions of the double-sided tool rotating structure **5**, the small pressing wheel structure **6** and the large pressing wheel structure **7** are controlled.

(23) The rotating chuck **8** includes a pushing arm **27**, a rotating seat **28**, a fixed mount **29**, a driving motor **30**, a rotating disc **31** and clamping jaws **32**. The pushing arm **27** is rotatably connected with the base **2** and located above the base **2**. The fixed mount **29** is arranged on one side of the base **2**. The rotating seat **28** is rotatably connected with the fixed mount **29** and located in the fixed mount **29**. The output end of the pushing arm **27** is hinged to the rotating seat **28**. The driving motor **30** is arranged in the rotating seat **28**. The rotating disc **31** is arranged at the output end of the driving motor **30**, and the clamping jaws **32** are arranged above the rotating disc **31**.

(24) In the embodiment, during clamping, the rotating disc **31** is obliquely arranged so that the wheel hub is convenient to be clamped. The clamping jaws **32** are configured for clamping the wheel hub. After clamping is completed, the pushing arm **27** pushes the rotating seat **28**, so that the rotating seat **28** is overturned upwards, and the rotating disc **31** and the wheel hub are kept horizontal. The driving motor **30** drives the clamping jaws **32** to rotate, so that the wheel hub rotates, and tire dismounting or tire dismounting is completed.

(25) The pushing arm **27** includes a second cylinder **33**, an ejector rod **34** and a first rotating rod **35**. The bottom of the second cylinder **33** is rotatably connected with the base **2**. The ejector rod **34** is arranged at the output end of the second cylinder **33**, and the first rotating rod **35** is rotatably connected with the ejector rod **34** and located at the end, away from the second cylinder **33**, of the ejector rod **34**.

(26) In the embodiment, the second cylinder **33** pushes the ejector rod **34** to move. In the moving process of the ejector rod **34**, due to the fact that the first rotating rod **35** is fixedly connected with the rotating seat **28**, the rotating seat **28** is overturned. Therefore, the inclination angle of the rotating seat **28** is adjusted.

(27) The rotating seat **28** includes a seat body **36**, a second rotating rod **37**, a frame body **38** and a fixed rod **39**. The two ends of the fixed rod **39** are fixedly connected with the fixed mount **29**. The second rotating rod **37** is rotatably connected with the fixed rod **39**. One end of the second rotating rod **37** is fixedly connected with the seat body **36**, and the other end of the second rotating rod **37** is fixedly connected with the frame body **38**. The frame body **38** is connected with the first rotating rod **35**.

(28) In the embodiment, the second cylinder **33** pushes the ejector rod **34** to move, and the frame body **38** rotates along with the first rotating rod **35**, so that the second rotating rod **37** rotates around the fixed rod **39**, and thus the seat body **36** rotates.

(29) When the vertical tire changer with a double-sided tool **17** and an overturning structure in the embodiment is used, the wheel hub is clamped above the rotating chuck **8**, and the rotating chuck **8** rotates to the horizontal position. A switch on the operating platform **3** is started for mounting or dismounting the tire. When the tire is mounted, the straight hook portion of the double-sided tool rotating structure **5** is located below the bent hook portion of the double-sided tool rotating

structure **5**. The operating platform **3** is configured for controlling the straight hook portion to walk to the position above a tire bead. Meanwhile, the small pressing wheel structure **6** is stretched to the position above the tire bead, and then the lifting seat is controlled to move downwards. In the process that the lifting seat moves downwards, the rotating chuck **8** drives the wheel hub to rotate by one circle to complete the mounting of a first crossing. The wheel hub continues to rotate downwards by one circle in the same method to complete the mounting of a second crossing. When the tire is dismounted, the small pressing wheel structure **6** is retracted, and the handball **15** is pulled upwards, so that the tool head **21** can freely rotate by 180°, and the bent hook portion is located below the straight hook portion. After the handball **15** is loosened, the locking pin shaft **16** falls into a round hole to achieve locking, and the bent hook portion hooks the tire. The rotating chuck **8** drives the wheel hub to rotate by one circle to complete the dismounting of the first crossing. Then, the large pressing wheel structure **7** walks to the edge of the wheel hub, the wheel hub rotates again, the tire is cut out of the edge of the wheel hub, and tire dismounting is completed.
(30) What is disclosed above is merely example embodiments of the present disclosure, and certainly is not intended to limit the protection scope of the present disclosure. Those skilled in the art may understand that all or some of processes that implement the foregoing embodiments and equivalent modifications made in accordance with the claims of the present disclosure shall fall within the scope of the present disclosure.

Claims

1. A vertical tire changer with a double-sided tool and an overturning structure, comprising a pedal, a base and an operating platform, the operating platform being arranged on a first side of the base, and the pedal being arranged beneath the operating platform, wherein the vertical tire changer further comprises a sliding body, a double-sided tool rotating structure, a small pressing wheel structure, a large pressing wheel structure, a lifting seat and a rotating chuck, the sliding body is arranged above the base, the lifting seat is arranged above the base, the sliding body is arranged on one side of the lifting seat, the double-sided tool rotating structure is mounted on one side of the sliding body pulley, the small pressing wheel structure and the large pressing wheel structure are mounted on an other side of the sliding body, the small pressing wheel structure is located above the large pressing wheel structure, the rotating chuck is arranged on a second side of the base, and the pedal is configured for controlling the rotating chuck.
2. The vertical tire changer with a double-sided tool and an overturning structure according to claim 1, wherein a first mounting hole is formed in the one side of the sliding body, and a second mounting hole and a third mounting hole are formed in the other side of the sliding body.
3. The vertical tire changer with a double-sided tool and an overturning structure according to claim 2, wherein the double-sided tool rotating structure comprises a first rod body, a cylinder lug, a pin seat, a handball, a locking pin shaft, a double-sided tool and a pin, the first rod body is mounted on the one side of the sliding body through the first mounting hole, the pin seat is arranged at one end of the first rod body, the locking pin shaft is arranged inside the pin seat, the handball is arranged on a top of the locking pin shaft, and the pin is arranged between the pin seat and the first rod body.
4. The vertical tire changer with a double-sided tool and an overturning structure according to claim 3, wherein the double-sided tool comprises a rotating shaft, set screws, a tool head and a hexagon socket head cap screw, the rotating shaft is arranged inside the pin seat, the tool head is detachably connected with the rotating shaft through the hexagon socket head cap screw, and the set screws are arranged between the tool head and the rotating shaft.
5. The vertical tire changer with a double-sided tool and an overturning structure according to claim 4, wherein the double-sided tool further comprises a first insert and a second insert, and the

first insert and the second insert are arranged at two ends of the tool head respectively.

6. The vertical tire changer with a double-sided tool and an overturning structure according to claim 5, wherein the small pressing wheel structure comprises a nylon wheel and a pressing rod, one end of the pressing rod is mounted on the one side of the sliding body through the second mounting hole, and the nylon wheel is mounted at an other end of the pressing rod.

7. The vertical tire changer with a double-sided tool and an overturning structure according to claim 6, wherein the lifting seat comprises a slideway and a cylinder rod, the slideway is fixedly connected with the base and located above the base, the sliding body is slidably connected with the slideway and located on one side of the slideway, the cylinder rod is arranged on an other side of the slideway, and the cylinder rod is connected with the sliding body.